

Dr. Joseph Karianjahi Njeri

Full-Stack Data Scientist | Backend Developer | Analytics Engineer

Python | Django | REST APIs | PostgreSQL | Machine Learning | JavaScript | Cloud Deployment

Contact Information

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GitHub	github.com/karianjahi
Portfolio	karianjahi.github.io/personal-portfolio

Professional Profile

Full-Stack Data Scientist and Software Developer with a PhD (Dr. rer. nat.) and several years of experience in analytics, backend development, machine learning, and scientific software engineering.

Specialized in designing AI-powered analytics platforms, operational intelligence systems, backend APIs, interactive dashboards, and machine learning pipelines using Python, Django, Django REST Framework, PostgreSQL, pandas, scikit-learn, and JavaScript.

Experienced in REST API development, data pipelines, KPI reporting, interactive dashboards, machine learning workflows, synthetic data generation, cloud deployment, and end-to-end SaaS application development.

Core Competencies

- **Backend Development:** Python, Django, Django REST Framework, REST APIs
- **Data Science & Analytics:** pandas, NumPy, scikit-learn, Machine Learning, Statistical Modeling
- **Databases:** PostgreSQL, SQL, Data Modeling
- **Frontend:** JavaScript, HTML, CSS, Chart.js
- **Deployment:** Railway, Render, Linux, Gunicorn, WhiteNoise
- **Software Engineering:** Git, Debugging, Refactoring, Testing, Clean Code
- **Cloud Technologies:** Docker, AWS Fundamentals, CI/CD Concepts

Technical Projects

1. AppPulse – Customer Analytics Platform for Jira Vendors

Built a SaaS-style analytics platform for monitoring customer engagement, feature adoption, retention, and churn risk for Atlassian Marketplace applications.

- Developed CSV ingestion and validation workflows.
- Implemented customer health scoring and churn prediction models.
- Built REST APIs and interactive dashboards using Chart.js.
- Added historical trend analysis, alerts, and reporting features.
- Deployed the platform on Railway using PostgreSQL.

Stack: Django, Django REST Framework, PostgreSQL, pandas, scikit-learn, JavaScript, Chart.js

GitHub: github.com/karianjahi/atlassian-app-analytics

Live Demo: customerhealth-atlassian-app-analytics.up.railway.app

2. FleetIQ – AI-Powered Fleet Operations and Maritime Intelligence Platform[3pt] Designed, developed, and deployed an enterprise-style maritime operations platform for fleet monitoring, operational analytics, telemetry processing, fleet health assessment, and machine-learning-based vessel risk prediction.

- Developed a full-stack operational dashboard featuring KPI cards, fleet health analytics, operational alert monitoring, and interactive Chart.js visualisations.
- Built REST APIs for fleet KPIs, operational alerts, voyage analytics, vessel health scoring, and vessel-level operational reporting using Django REST Framework.
- Implemented a telemetry-driven operational alert engine capable of detecting engine overheat, fuel anomalies, weather risks, delay risks, and speed anomalies.
- Designed vessel detail pages with voyage history, operational KPIs, alert management, pagination, sorting, filtering, and health scoring.
- Implemented interactive dashboard drill-down navigation enabling exploration from fleet-level analytics to vessel-specific operational views.
- Developed a machine-learning feature engineering pipeline and Random Forest training workflow for vessel risk prediction using pandas and scikit-learn.
- Created a synthetic data generation engine capable of producing realistic vessels, voyages, telemetry records, operational alerts, and configurable fleet health profiles.
- Deployed the platform to Railway using PostgreSQL, Gunicorn, WhiteNoise, and production-ready Django configuration.

Stack: Django, Django REST Framework, PostgreSQL, pandas, scikit-learn, JavaScript, Chart.js, Railway

GitHub: github.com/karianjahi/fleetiq

Live Demo: fleetiq-maritime.up.railway.app

Professional Experience

08/2025 – Present Independent Full-Stack Development & Applied Machine Learning

Karlsruhe, Germany

- Expanded backend and full-stack engineering skills to complement prior machine learning and analytics experience
- Designed and developed deployed business intelligence and analytics applications using Django, PostgreSQL, REST APIs, and JavaScript
- Built SaaS-style analytics platforms for customer intelligence, KPI reporting, and churn-risk analysis
- Continuously improving the platforms with additional analytics, automation, and reporting features

11/2021 – 07/2025 Data Science Coach / Software Engineering Coach

Spiced Academy / neufische GmbH, Berlin, Germany

- Taught Python, SQL, Machine Learning, APIs, and software engineering
- Mentored practical full-stack and data science projects
- Supported learners with debugging, architecture, and deployment
- Introduced Git workflows and clean code practices

07/2019 – 06/2021 Data Scientist

Engineering Data Intelligence (EDI) GmbH, Karlsruhe, Germany

- Developed analytical pipelines and data-driven applications
- Processed and modeled large datasets
- Built machine learning and statistical models

05/2016 – 06/2019 Research Associate

Karlsruhe Institute of Technology (KIT), Germany

- Maintaining and updating a database for station meteorological data using python scripts

01/2013 – 06/2015 Lecturer

South Eastern Kenya University and University of Eldoret, Kenya

- Taught meteorology and environmental sciences
- Supervised student projects

Education**06/2021 – 09/2021 Data Science and Machine Learning Bootcamp**

Spiced Academy, Berlin, Germany

10/2015 – 05/2019 Dr. rer. nat. (PhD) in Meteorology

Karlsruhe Institute of Technology (KIT), Germany

Dissertation: Synoptic-scale dynamics and modeling of extreme precipitation in Kenya.

10/2010 – 08/2012 M.Sc. in Meteorology

University of Nairobi, Kenya

Languages

- English – Full Professional Proficiency (C2)
- German – Intermediate (B1/B2)
- Swahili – Native

Awards

- DAAD Scholarship

Professional Interests

Data Science, Analytics Engineering, Machine Learning, Backend Development, SaaS Analytics, Full-Stack Web Development, and Cloud Technologies.